

Assessment of factor IX activity of eftrenonacog alpha using one-stage clotting assays

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Conclusion

A 2-fold underestimation of rFIXFc biases factor IX assays using STA-CK Prest® as activator and phospholipid reagents.

Eftrenonacog alpha is a long-acting pharmaceutical product used as replacement therapy for patients suffering from haemophilia B. This study aimed to assess the performance of the methods used in the Belgian clinical laboratories for measuring factor IX activity in samples containing Recombinant Factor IX Fc Fusion Protein (rFIXFc).

Methods

- Samples were collected from a patient with haemophilia B treated with eftrenonacog alpha.
- They were distributed and lyophilized for the Belgian EQA scheme organized by Sciensano as part of the external quality assessment scheme of October 2018 for blood coagulation.
- Participants were requested to measure factor IX activity according to their usual procedures and to record the reagents and analyser used.
- Statistical analysis was performed by comparing methods using a Student test for groups with unequal variability. The P-values have been corrected for simultaneous hypothesis tests based on the Bonferroni correction.

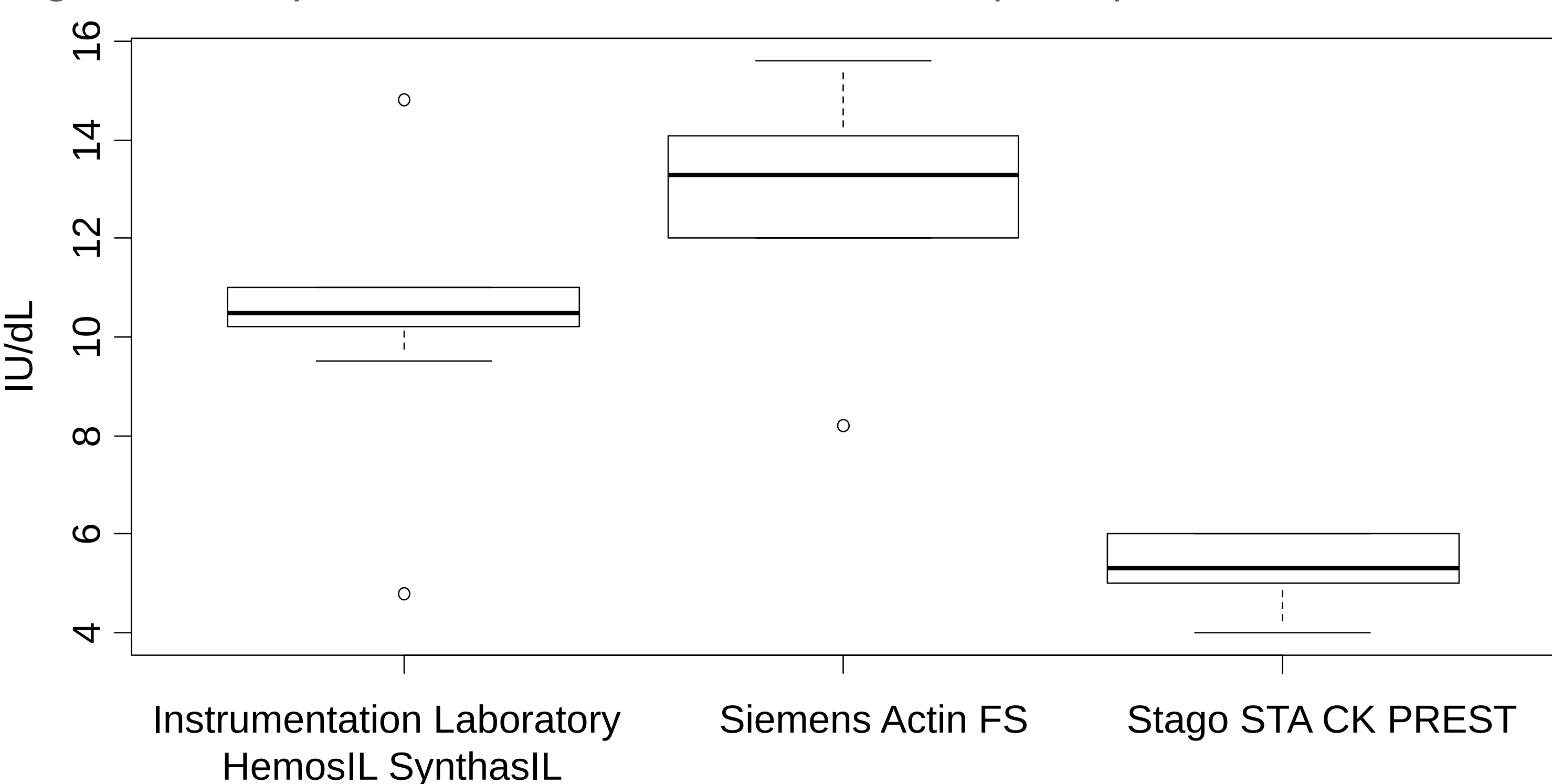
Results

- Twenty-one Belgian clinical laboratories participated in this study.
- All used a one-stage aPTT-based clotting assay.
- The results are summarized in Table 1.
- The difference between the results of the methods with at least 5 participants is illustrated in Figure 1.

Table 1. Summary of factor IX assay results. Method-specific medians (IU/dL) and coefficients of variation (CV, %) are shown for all methods with ≥5 reported results. Results (IU/dL) are presented individually for the other methods

Method	Median/ result	CV	Range	N
Instrumentation Laboratory HemosIL SynthasIL	10.5	5.6	4.8 - 14.8	9
Siemens Actin	16.0			1
Siemens Actin FS	13.3	11.6	8.2 – 15.6	5
Stago STA CK PREST	5.3	14.0	4.0 – 6.0	5
Stago STA-Cephascreen	4.2			1

Figure 1. Box plots of the methods with at least 5 participants



- A statistical comparison could be made between the three methods with at least 5 participants. The results are shown in table 2.
- The results obtained with STA-CK Prest® are significantly lower than those obtained with Actin FS® and SynthASil®.
- There is no statistically significant difference between Actin FS® and SynthASil®.

Table 2. Comparison of the methods with at least 5 participants

Comparison	Difference	P-value
Instrumentation Laboratory HemosIL SynthasIL - Siemens Actin FS	-2.336	0.4893
Instrumentation Laboratory HemosIL SynthasIL - Stago STA CK PREST	5.04	0.000727
Siemens Actin FS - Stago STA CK PREST	7.376	0.008874

Discussion

- With the exception of the single observation with STA-Cephascreen, these results are in agreement with published data. According to Sommer et al¹, accurate measurements of rFIXFc can be achieved with SynthasIL, Actin, Actin FS and STA-Cephascreen. By contrast, a 2-fold underestimation of rFIXFc biases factor IX assays using STA-CK-PREST.
- The strong point of our study is the use of a sample collected from a patient. However, the limited quantity of sample made it impossible to include all Belgian laboratories and the number of results was limited for certain methods. Another limitation is that we investigated the performance of the assays at only one concentration.

REFERENCES

1. Sommer JM, Buyue Y, Bardan S, et al. Comparative field study: impact of laboratory assay variability on the assessment of recombinant factor IX Fc fusion protein (rFIXFc) activity. *Thromb Haemost.* 2014 Nov;112(5):932-40